**Linear Regression with multiple variables**

1

**Multiple features (variables)**

| Size (feet²) | Price ($1000) |
|---|---|
| $x$ | $y$ |
| 2104 | 460 |
| 1416 | 232 |
| 1534 | 315 |
| 852 | 178 |
| ... | ... |

$$h_\theta(x) = \theta_0 + \theta_1 x$$

2

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

3

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$n$ = number of features
$x^{(i)}$ = input (features)of $i^{th}$ training example
$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

4

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$n$ = number of features → 4
$x^{(i)}$ = input (features) of $i^{th}$ training example
$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

5

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$i = 2$

$n$ = number of features → 4
$x^{(i)}$ = input (features) of $i^{th}$ training example
$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

6

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$x^{(2)}$

$n$ = number of features → 4
$x^{(i)}$ = input (features) of $i^{th}$ training example
$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

7

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$n$ = number of features → 4
$x^{(i)}$ = input (features) of $i^{th}$ training example
$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

$$x^{(2)} = \begin{bmatrix} 1416 \\ 3 \\ 2 \\ 40 \end{bmatrix}$$

8

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$n$ = number of features → 4

$x^{(i)}$ = input (features) of $i^{th}$ training example

$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

$j=3$

$$x^{(2)} = \begin{bmatrix} 1416 \\ 3 \\ 2 \\ 40 \end{bmatrix}$$

9

**Multiple features (variables)**

| Size (feet²) | Number of bedrooms | Number of floors | Age of home (years) | Price ($1000) |
|---|---|---|---|---|
| 2104 | 5 | 1 | 45 | 460 |
| 1416 | 3 | 2 | 40 | 232 |
| 1534 | 3 | 2 | 30 | 315 |
| 852 | 2 | 1 | 36 | 178 |
| ... | ... | ... | ... | ... |

$n$ = number of features → 4

$x^{(i)}$ = input (features) of $i^{th}$ training example

$x_j^{(i)}$ = value of feature $j$ in $i^{th}$ *training example*

$$x^{(2)} = \begin{bmatrix} 1416 \\ 3 \\ 2 \\ 40 \end{bmatrix}$$

$x_3^2 = 2$

10

**Hypothesis:**

**Previously:** $h_\theta(x) = \theta_0 + \theta_1 x$

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**Hypothesis:**

**Previously:** ~~$h_\theta(x) = \theta_0 + \theta_1 x$~~

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

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**Hypothesis:**

Previously: $h_\theta(x) = \theta_0 + \theta_1 x$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$e.g.$ $h_\theta(x) = 80 + 0.1x_1 + 0.01x_2 + 3x_3 - 2x_4$

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**Hypothesis for multivariables**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

14

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1$

15

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1$ $\Longrightarrow (x_0^{(i)} = 1)$

16

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1 \quad \Rightarrow (x_0^{(i)} = 1)$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

17

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1 \quad \Rightarrow (x_0^{(i)} = 1)$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \cdot \\ \cdot \\ \cdot \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

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**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1 \quad \Rightarrow (x_0^{(i)} = 1)$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \cdot \\ \cdot \\ \cdot \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$
$$= \theta^T x$$

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**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \cdot \\ \cdot \\ \cdot \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$[\theta_0 \ \theta_1 \ \theta_2 \ \dots \theta_n] \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix}$$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$
$$= \theta^T x$$

20

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \cdot \\ \cdot \\ \cdot \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$[\theta_0\ \theta_1\ \theta_2\ \dots \theta_n] \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \Rightarrow \theta^T x$$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$
$$= \theta^T x$$

21

**Hypothesis for multivariable**

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$

For convenience of notation, define $x_0 = 1$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \cdot \\ \cdot \\ \cdot \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$[\theta_0\ \theta_1\ \theta_2\ \dots \theta_n] \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \Rightarrow \theta^T x$$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$$
$$= \theta^T x$$

**Multi-variate Linear Regression**

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**Hypothesis:** $h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$

**Parameters:** $\theta_0, \theta_1, \theta_2, \dots, \theta_n$

**Cost Function:** $J(\theta_0, \theta_1, \theta_2, \dots \theta_n) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^i) - y^i)^2$

23

**Hypothesis:** $h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$

**Parameters:** $\theta_0, \theta_1, \theta_2, \dots, \theta_n$

**Cost Function:** $J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^i) - y^i)^2$

24

**Hypothesis:** $h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$

**Parameters:** $\theta_0, \theta_1, \theta_2, \dots, \theta_n$

**Cost Function:** $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$

**Gradient descent:**

**Repeat** {

$\boldsymbol{\theta_j} := \boldsymbol{\theta_j} - \boldsymbol{\alpha}\ \frac{\partial}{\partial \theta_j} J(\ \theta_0, \theta_1, \theta_2, \dots, \theta_n)$

} (Simultaneously update for every $\boldsymbol{j = 0, \dots, n}$)

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**Hypothesis:** $h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \dots \ + \theta_n x_n$

**Parameters:** $\theta_0, \theta_1, \theta_2, \dots, \theta_n$

**Cost Function:** $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$

**Gradient descent:**

**Repeat** {

$\boldsymbol{\theta_j} := \boldsymbol{\theta_j} - \boldsymbol{\alpha}\ \frac{\partial}{\partial \theta_j} ~~J(\ \theta_0, \theta_1, \theta_2, \dots, \theta_n)~~ \rightarrow J(\theta)$

} (Simultaneously update for every $\boldsymbol{j = 0, \dots, n}$)

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**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})\ .\ x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

27

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})\ .\ x^{(i)}$

$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$

$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

28

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

**New algorithm** ($n \geq 1$)

Repeat until convergence {

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

29

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

**New algorithm** ($n \geq 1$)

Repeat until convergence {

$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

30

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

**New algorithm** ($n \geq 1$)

Repeat until convergence {

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

$\boldsymbol{\theta_0} \longrightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$

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**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0, \theta_1}$)
}

**New algorithm** ($n \geq 1$)

Repeat until convergence {

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

$\boldsymbol{\theta_0} \longrightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$

32

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0}, \boldsymbol{\theta_1}$)
}

**New algorithm** $(n \geq 1)$

Repeat until convergence {

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

$\boldsymbol{\theta_0} \longrightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$

$\boldsymbol{\theta_1} \longrightarrow \theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_1^{(i)}$

33

**Gradient descent algorithm**

Previously (n=1)

Repeat until convergence {

$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$

$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_0}, \boldsymbol{\theta_1}$)
}

**New algorithm** $(n \geq 1)$

Repeat until convergence {

$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

(simultaneously update for every $\boldsymbol{\theta_j}$ for $\boldsymbol{j = 0, \ldots, n}$)
}

$\boldsymbol{\theta_0} \longrightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$

$\boldsymbol{\theta_1} \longrightarrow \theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_1^{(i)}$

$\boldsymbol{\theta_2} \longrightarrow \theta_2 := \theta_2 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_2^{(i)}$

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## Linear Regression with multiple variables

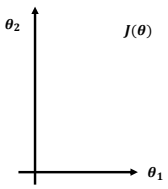
Gradient descent in practice: Features Scaling

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**Feature Scaling**

Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)

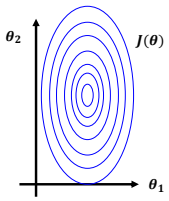



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**Feature Scaling**

Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)

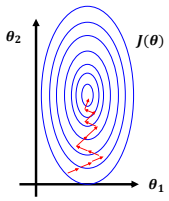



37

**Feature Scaling**

Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)

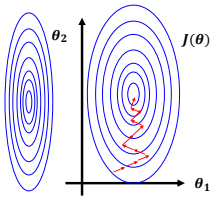



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**Feature Scaling**

Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)

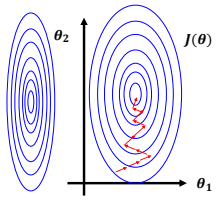



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**Feature Scaling**

Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)




$$x_1 = \frac{\text{size(feet}^2)}{2000}$$

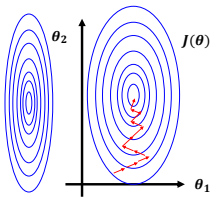
$$x_2 = \frac{\text{number of bedrooms}}{5}$$

40

**Feature Scaling**

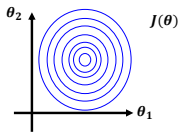
Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)




$x_1 = \frac{\text{size(feet}^2)}{2000}$

$x_2 = \frac{\text{number of bedrooms}}{5}$

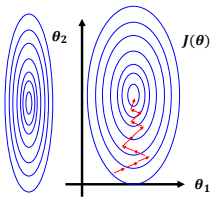



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**Feature Scaling**

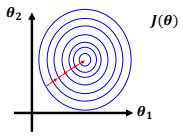
Idea: Make sure features are on a similar scale

e.g. $x_1$ = size(0 – 2000 feet$^2$)
$x_2$ = numer of bedrooms (1 – 5)




$x_1 = \frac{\text{size(feet}^2)}{2000}$

$x_2 = \frac{\text{number of bedrooms}}{5}$




42

**Feature Scaling**

Get every feature into approximately a $-1 \leq x_i \leq 1$ range.

$x_0 = 1$

e.g. $0 \leq x_1 \leq 3$

$-2 \leq x_2 \leq 0.5$

$-100 \leq x_3 \leq 100$

$-0.0001 \leq x_4 \leq 0.0001$

43

**Feature Scaling**

Get every feature into approximately a $-1 \leq x_i \leq 1$ range.

$x_0 = 1$

e.g. $0 \leq x_1 \leq 3$ ✓

$-2 \leq x_2 \leq 0.5$ ✓

$-100 \leq x_3 \leq 100$ ✗

$-0.0001 \leq x_4 \leq 0.0001$ ✗

44

**Feature Scaling**

Get every feature into approximately a $-1 \leq x_i \leq 1$ range.

$x_0 = 1$

e.g. $0 \leq x_1 \leq 3$ ✓

$-2 \leq x_2 \leq 0.5$ ✓

$-100 \leq x_3 \leq 100$ ✗

$-0.0001 \leq x_4 \leq 0.0001$ ✗

$-3 \leq x_1 \leq 3$

$-1/3 \leq x_1 \leq 1/3$

45

**Feature Scaling**

Replace $x_i$ with $x_i - \mu_i$ to make features have approximately zero mean (Do not apply to $x_0$ )

46

**Feature Scaling**

Replace $x_i$ with $x_i - \mu_i$ to make features have approximately zero mean (Do not apply to $x_0$ )

$$x_1 = \frac{size - 1000}{2000}$$

$$x_2 = \frac{\#bedrooms - 2}{5}$$

47

**Feature Scaling**

Replace $x_i$ with $x_i - \mu_i$ to make features have approximately zero mean (Do not apply to $x_0$ )

$$x_1 = \frac{size - 1000}{2000}$$

$$x_2 = \frac{\#bedrooms - 2}{5}$$

Generally,

$$x_i = \frac{x_i - \mu_i}{s_i}$$

48

**Feature Scaling**

Replace $x_i$ with $x_i - \mu_i$ to make features have approximately zero mean
(Do not apply to $x_0$ )

$$x_1 = \frac{size - 1000}{2000}$$

$$x_2 = \frac{\#bedrooms - 2}{5}$$

Generally,

$$x_i = \frac{x_i - \mu_i}{s_i}$$

Average value of $x_i$ in the training set

49

**Feature Scaling**

Replace $x_i$ with $x_i - \mu_i$ to make features have approximately zero mean
(Do not apply to $x_0$ )

$$x_1 = \frac{size - 1000}{2000}$$

$$x_2 = \frac{\#bedrooms - 2}{5}$$

Generally,

$$x_i = \frac{x_i - \mu_i}{s_i}$$

Average value of $x_i$ in the training set

range of $x_i$ in the training set
(max – min) or (std. deviation)

50

**Is gradient descent working correctly? Is the learning rate appropriate?**

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

51

**Is gradient descent working correctly? Is the learning rate appropriate?**

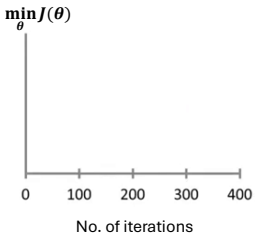
$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

- "Debugging": How to make sure gradient descent is working correctly

- How to choose learning rate $\alpha$

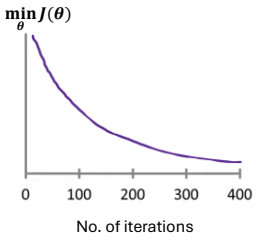
52

**Is gradient descent working correctly?**



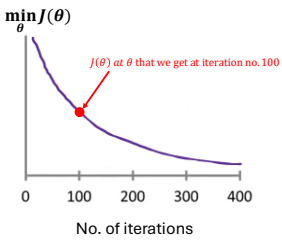

53

**Is gradient descent working correctly?**



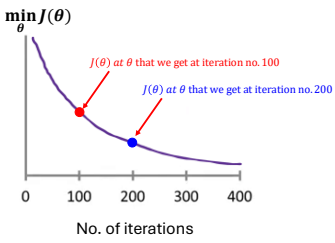

54

**Is gradient descent working correctly?**



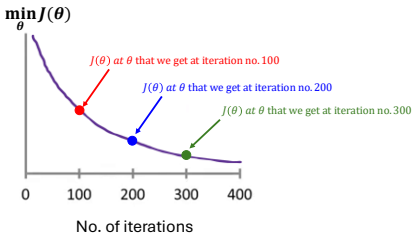

55

**Is gradient descent working correctly?**



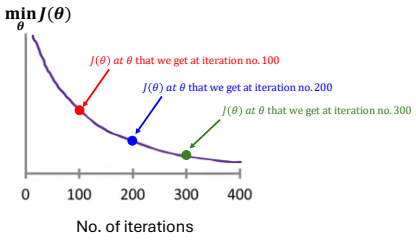

56

**Is gradient descent working correctly?**




57

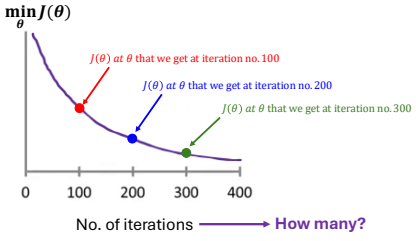
**Is gradient descent working correctly?**




**Conclusion**: If gradient descent is working properly, then $J(\theta)$ must **decrease** after every iteration

58

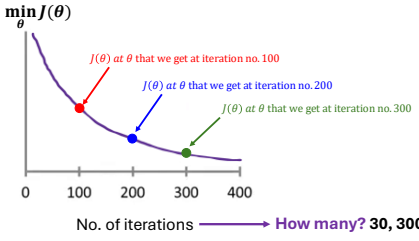
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**Conclusion**: If gradient descent is working properly, then $J(\theta)$ must **decrease** after every iteration

59

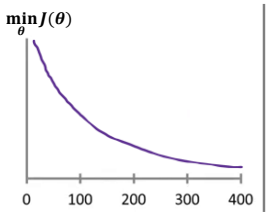
**Is gradient descent working correctly?**




**Conclusion**: If gradient descent is working properly, then $J(\theta)$ must **decrease** after every iteration

60

**Is gradient descent working correctly?**




**Automatic Convergence test:**

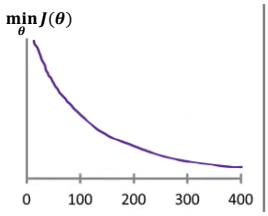
Declare convergence if $J(\theta)$ decreases by less than $10^{-3}$ in one iteration

No. of iterations → **How many? 30, 3000, 300,000 etc**

**Conclusion**: If gradient descent is working properly, then $J(\theta)$ must **decrease** after every iteration

61

**Is gradient descent working correctly?**




**Automatic Convergence test:**

Declare convergence if $J(\theta)$ decreases by less than $10^{-3}$ in one iteration

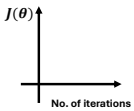
Setting this threshold is often **DIFFICULT**

No. of iterations → **How many? 30, 3000, 300,000 etc**

**Conclusion**: If gradient descent is working properly, then $J(\theta)$ must **decrease** after every iteration

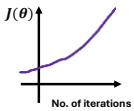
62

**Is gradient descent working correctly?**



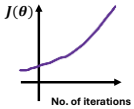

63

**Is gradient descent working correctly?**

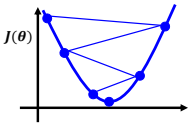



64

**Is gradient descent working correctly?**

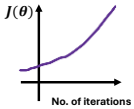



Gradient descent is not working properly
Use smaller **LEARNING RATE (alpha)**

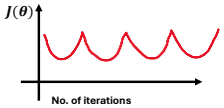


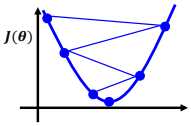

65

**Is gradient descent working correctly?**



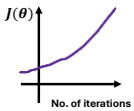

Gradient descent is not working properly
Use smaller **LEARNING RATE (alpha)**



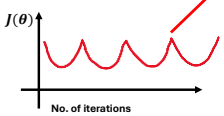



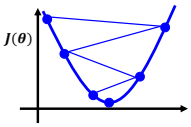

66

**Is gradient descent working correctly?**



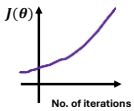

Gradient descent is not working properly
Use smaller **LEARNING RATE (alpha)**



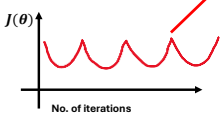



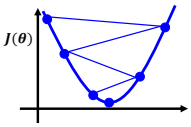

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**Is gradient descent working correctly?**




Gradient descent is not working properly
Use smaller **LEARNING RATE (alpha)**




**Conclusion:**
- For sufficiently small ALPHA, $J(\theta)$ should decrease on every iteration.
- But if ALPHA is too small, gradient descent can be slow to converge.

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**Summary:**

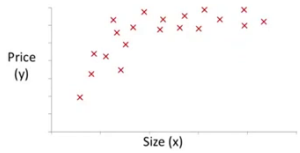
- If ALPHA is too SMALL: Slow convergence
- If ALPHA is too LARGE: $J(\theta)$ may not decrease on every iteration; may not converge

To choose ALPHA, try

…,0.0001, ,0.01, 0.1, 1, …

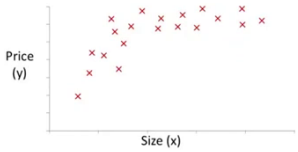
69

**Polynomial regression**




70

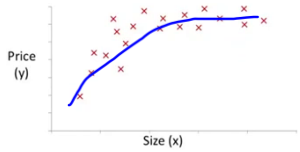
**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

71

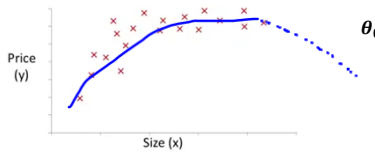
**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

72

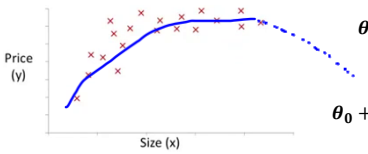
**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

73

**Polynomial regression**

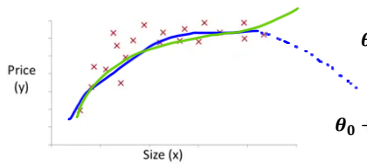



$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

74

**Polynomial regression**

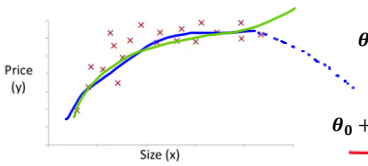



$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

75

**Polynomial regression**



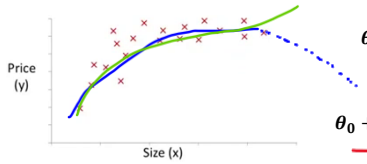

$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

how??

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**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

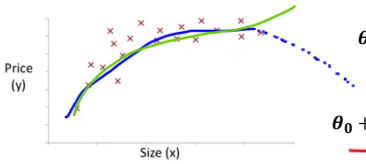
how??

$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

$x_1 = size$
$x_2 = (size)^2$
$x_3 = (size)^3$

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**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

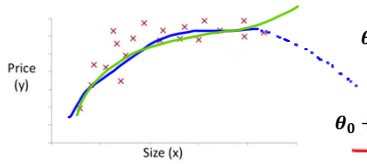
$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

how??

$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$
$= \theta_0 + \theta_1 (size) + \theta_2 (size)^2 + \theta_3 (size)^3$

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**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

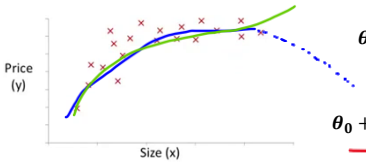
how??

$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$
$= \theta_0 + \theta_1 (size) + \theta_2 (size)^2 + \theta_3 (size)^3$

$x_1 = size$
$x_2 = (size)^2$
$x_3 = (size)^3$

79

**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

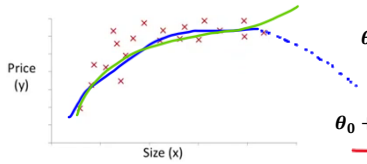
how??

$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$
$= \theta_0 + \theta_1 (size) + \theta_2 (size)^2 + \theta_3 (size)^3$

$x_1 = size$ → 1 – 1000
$x_2 = (size)^2$ → 1 – 1000,000
$x_3 = (size)^3$ → 1 – 1000,000,000

80

**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

how??

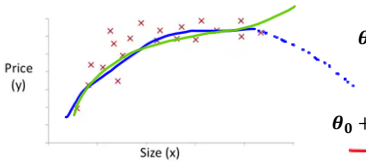
$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$
$$= \theta_0 + \theta_1(size) + \theta_2(size)^2 + \theta_3(size)^3$$

$x_1 = size$ → 1 – 1000
$x_2 = (size)^2$ → 1 – 1000,000
$x_3 = (size)^3$ → 1 – 1000,000,000

Problem???

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**Polynomial regression**




$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

how??

$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$
$$= \theta_0 + \theta_1(size) + \theta_2(size)^2 + \theta_3(size)^3$$

$x_1 = size$ → 1 – 1000
$x_2 = (size)^2$ → 1 – 1000,000
$x_3 = (size)^3$ → 1 – 1000,000,000

Feature Scaling

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